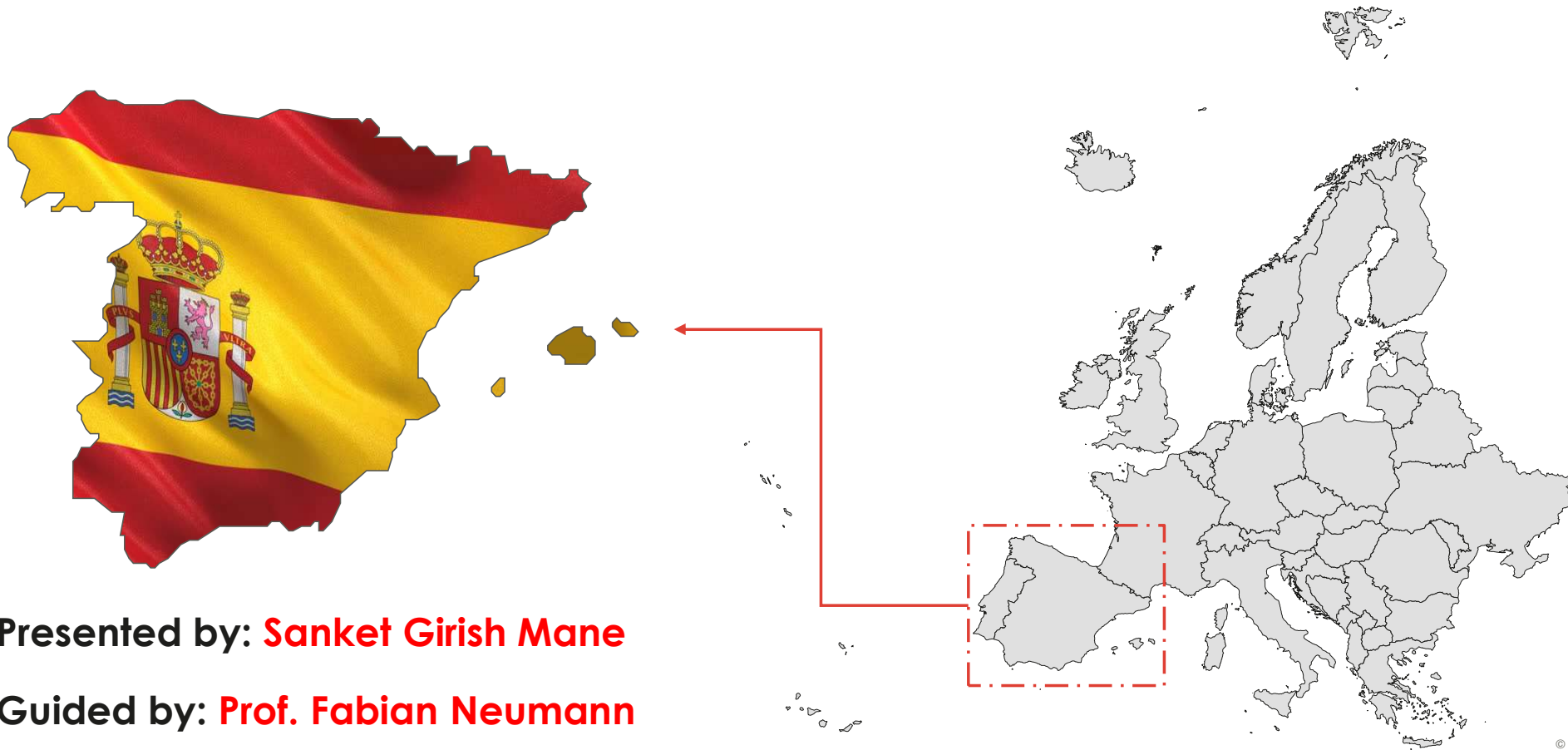


DATA SCIENCE FOR ENERGY SYSTEM MODELLING

FINAL CASE STUDY PRESENTATION



Presented by: **Sanket Girish Mane**

Guided by: **Prof. Fabian Neumann**

Agenda

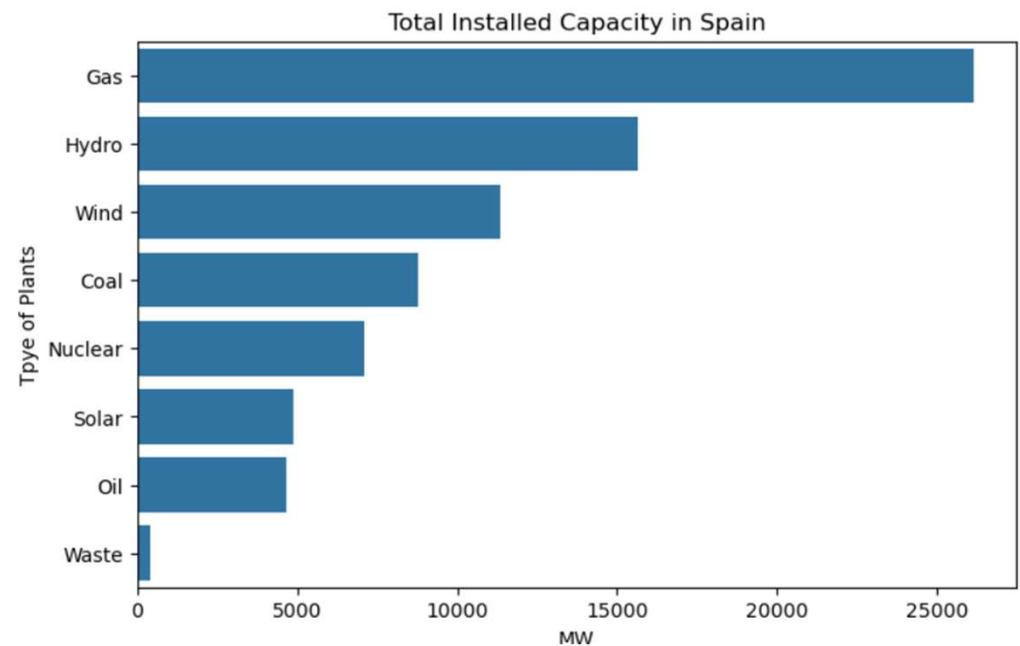
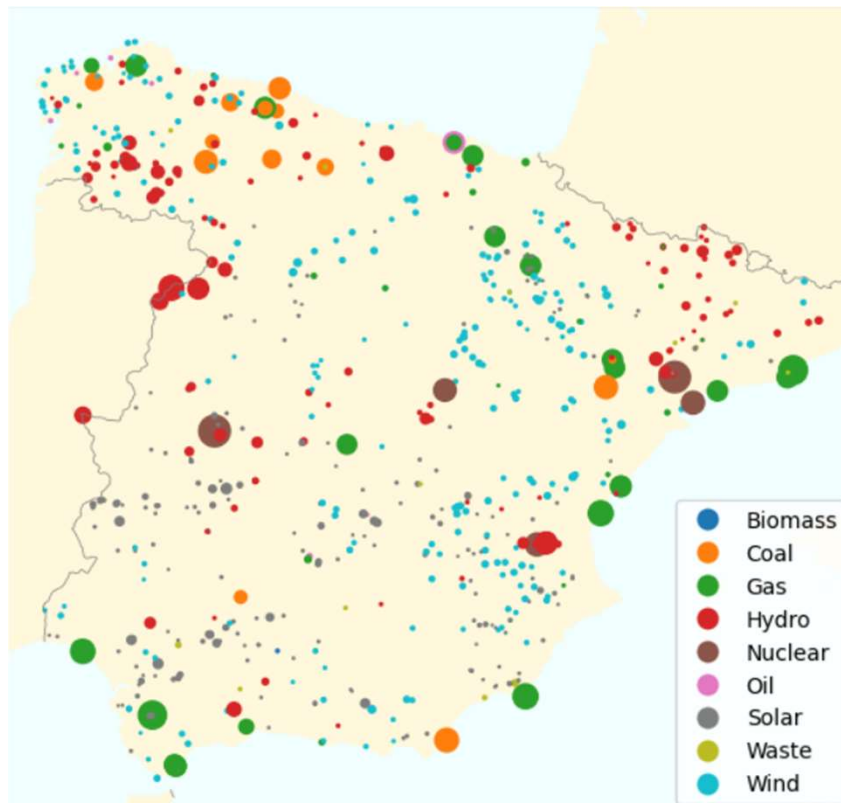
- Input Data
 - Existing Powerplants
 - Defined REGIONS and it's OCEAN share
 - Electricity demand per region
 - Land Eligibility Analysis
- Optimization Results
 - Base Scenario vs No CO₂ emissions vs Purely RES
- Sensitivity Analysis
 - Solar Dispatch and Cost Sensitivity
- Conclusions
- Future Outlook

INPUT DATA

- Existing Powerplants
- Defined REGIONS and it's OCEAN share
- Electricity demand per region
- Land Eligibility Analysis

Existing Powerplants

Input Data

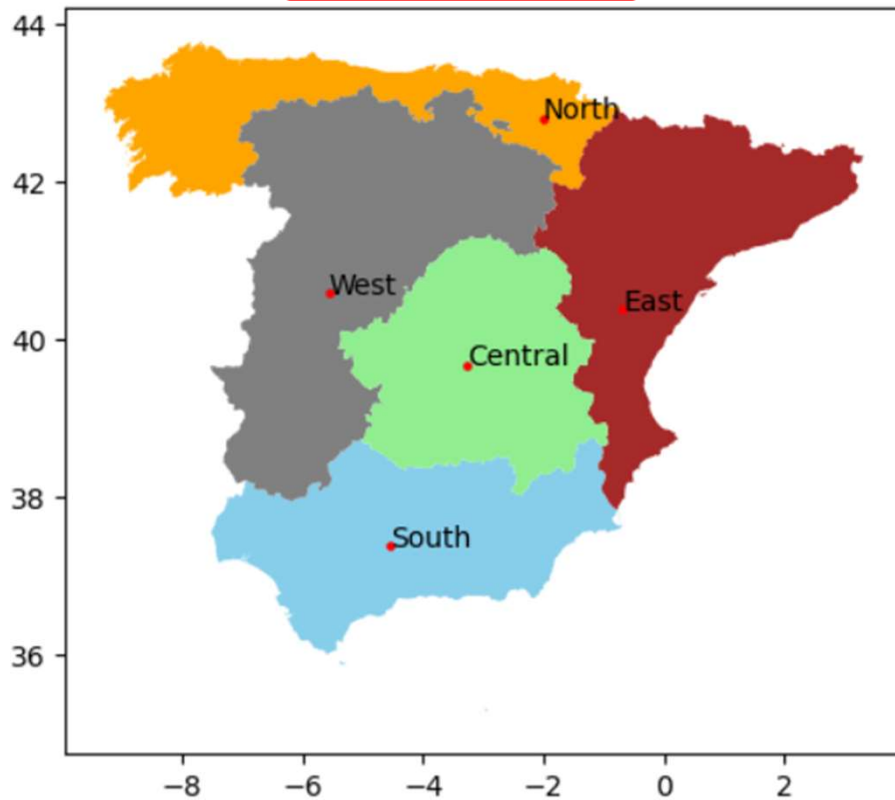


The plot and bar graph clearly shows that Spain is highly dependent upon **'Gas'**

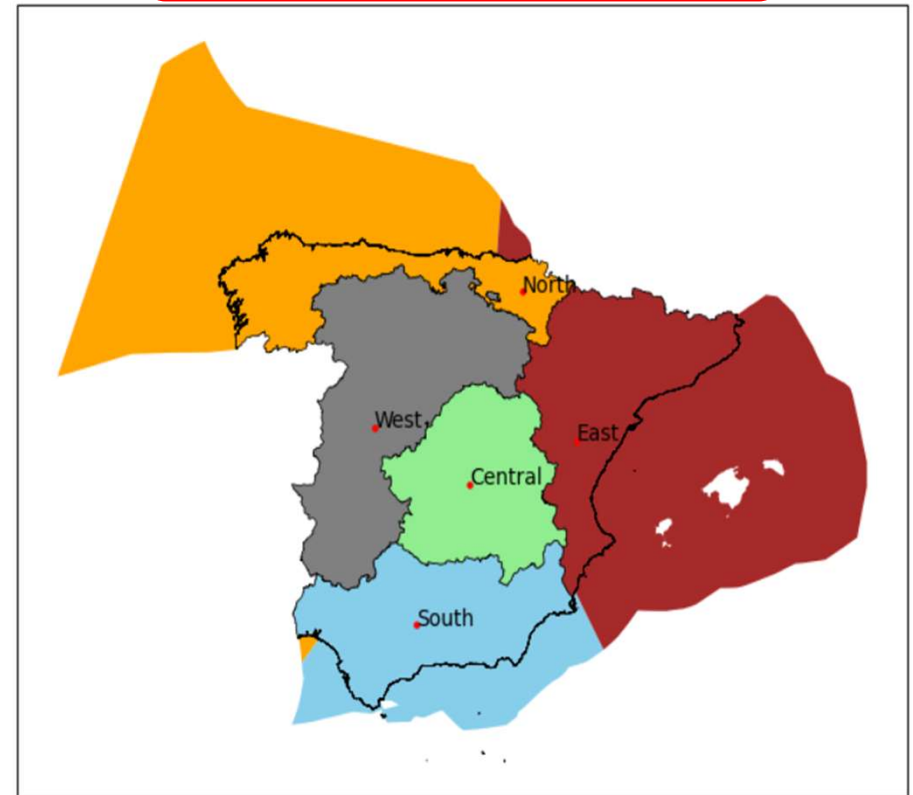
Defined REGIONS and it's OCEAN share

Input Data

LAND Regions

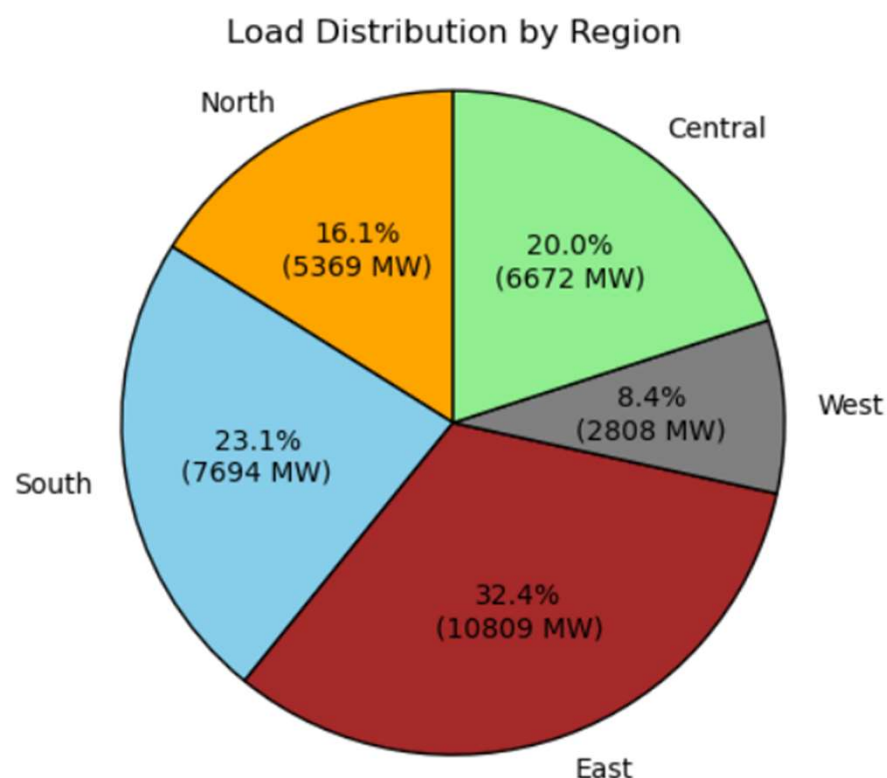


Defined REGIONS along with
it's OCEAN share



Load distribution by Region

Input Data



- Load distribution by Region was calculated based upon the population observed across different provenience of Spain^[1].
- As the Eastern part of Spain is highly populated, the electricity demand observed for this region is the highest.
- However, this region is highly depended upon big gas plant. (Ref. slide 3)
- Thus, based upon the existing scenario, Spain has very good opportunity to installed more Renewable Energy sources and connect it to the grid.

1. <https://www.statista.com/statistics/445549/population-of-spain-by-autonomous-community/>

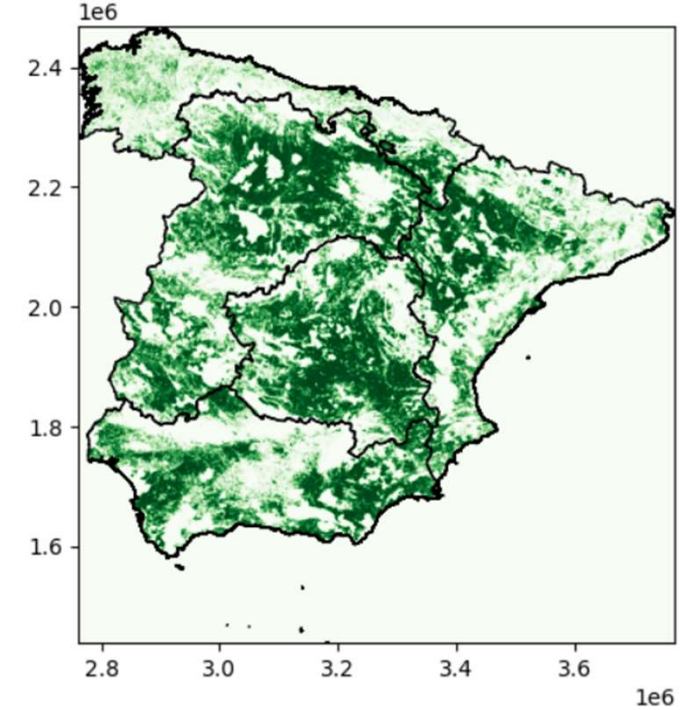
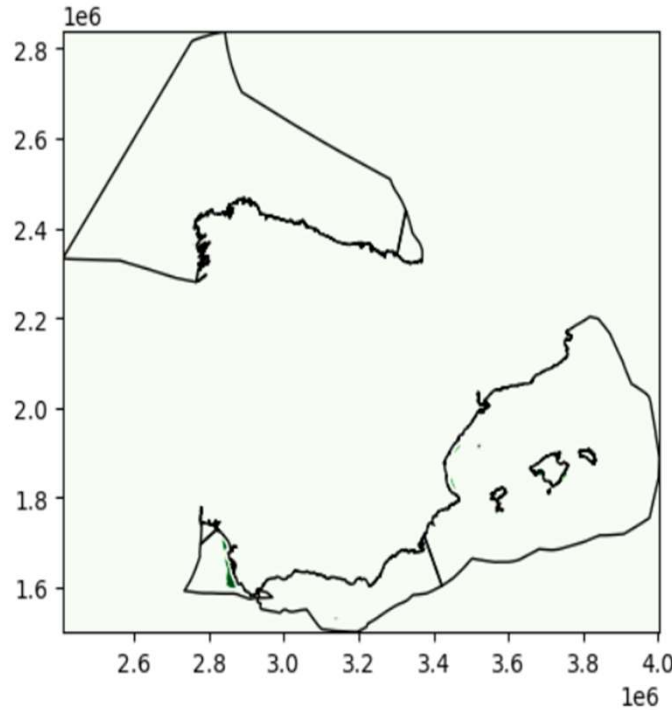
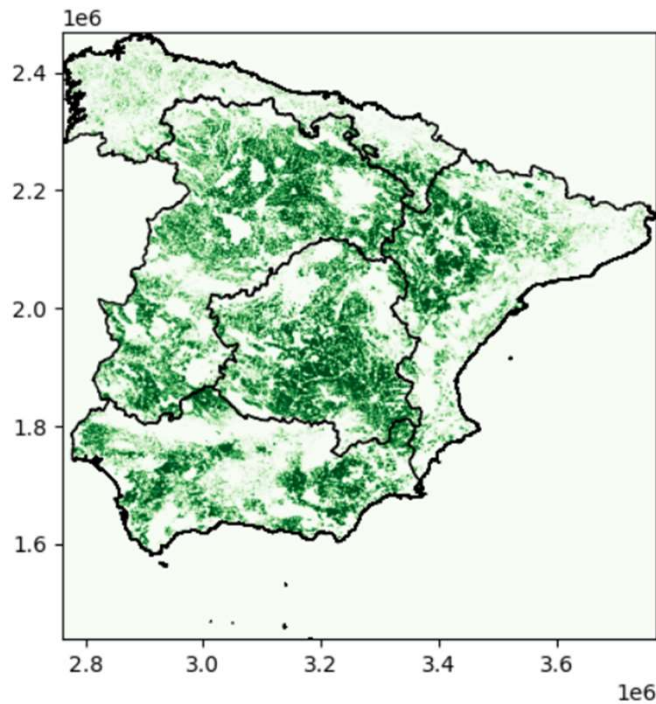
Land Eligibility Analysis

Input Data

For Onshore, **153944.2 km²** is available, which is **31.19%** of Spanish Land

For Offshore, **1582.5 km²** is available, which is **0.28%** of Spanish Land

For Solar, **209014.7 km²** is available, which is **42.35%** of Spanish Land



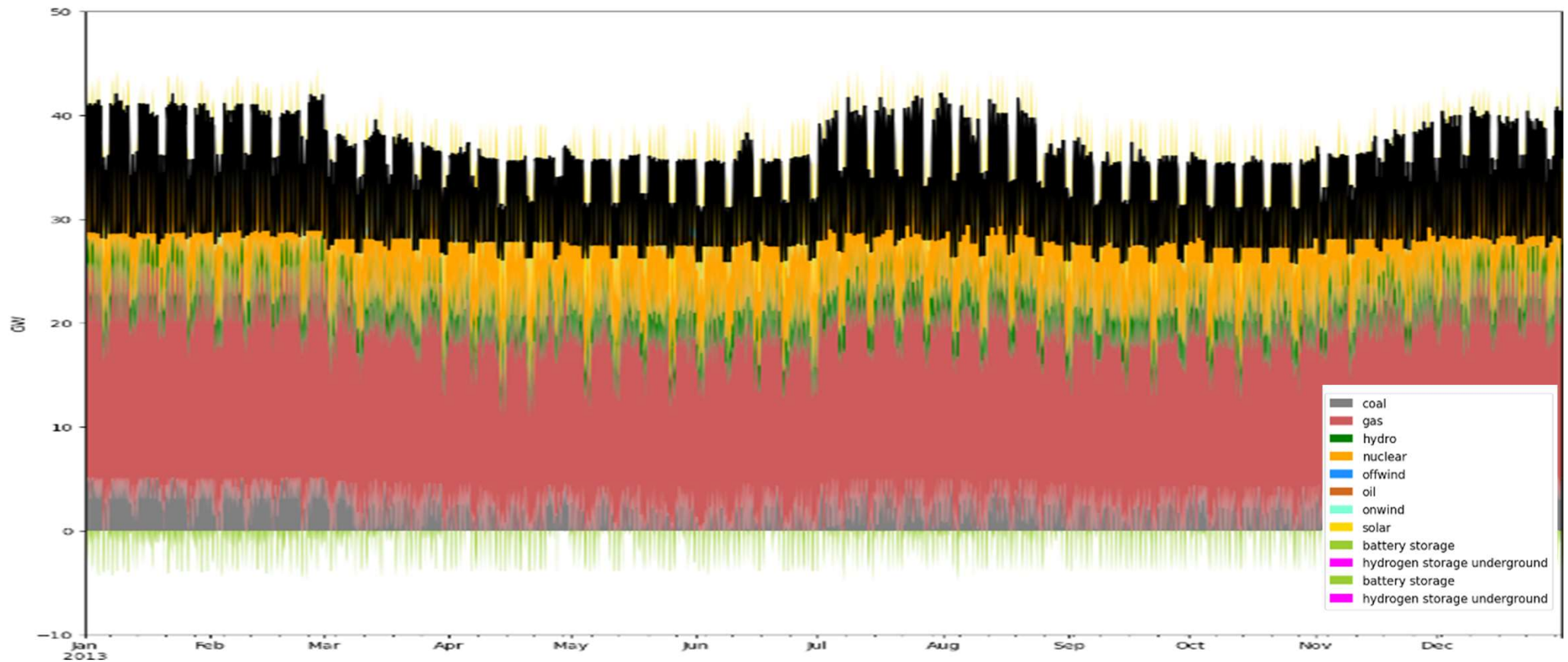
Note; Did not considered extra CORINE class codes

OPTIMIZATION RESULTS

- Base Scenario vs No CO₂ emissions vs Purely RES

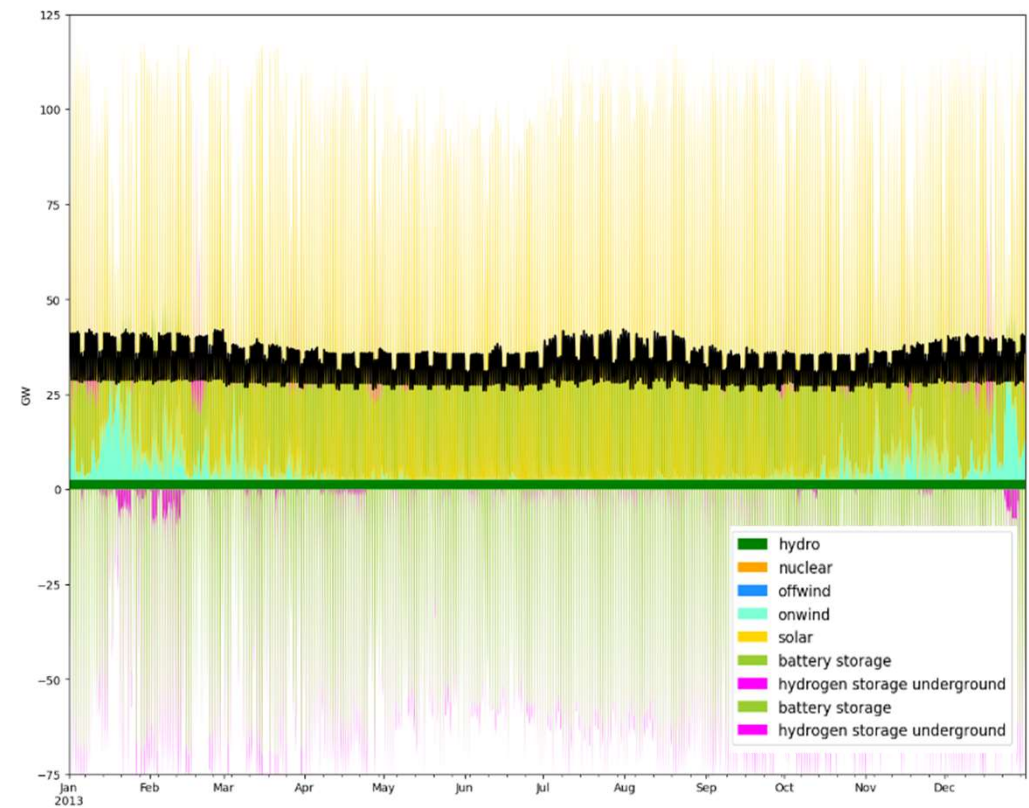
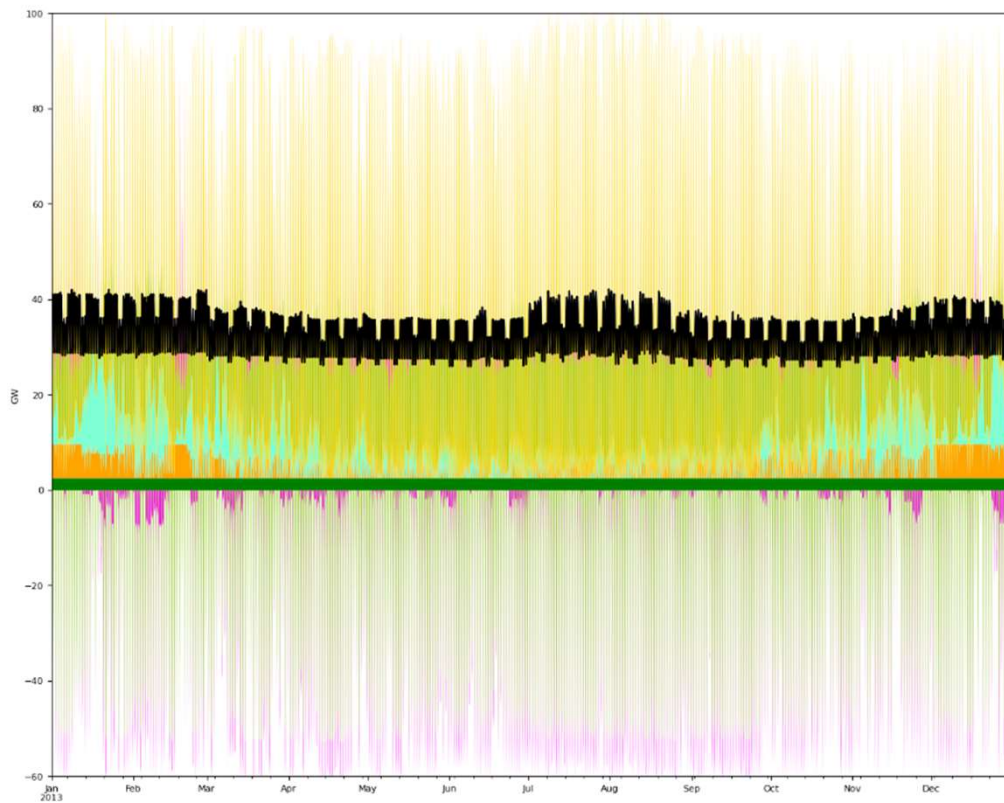
Dispatch Results

Base Scenario



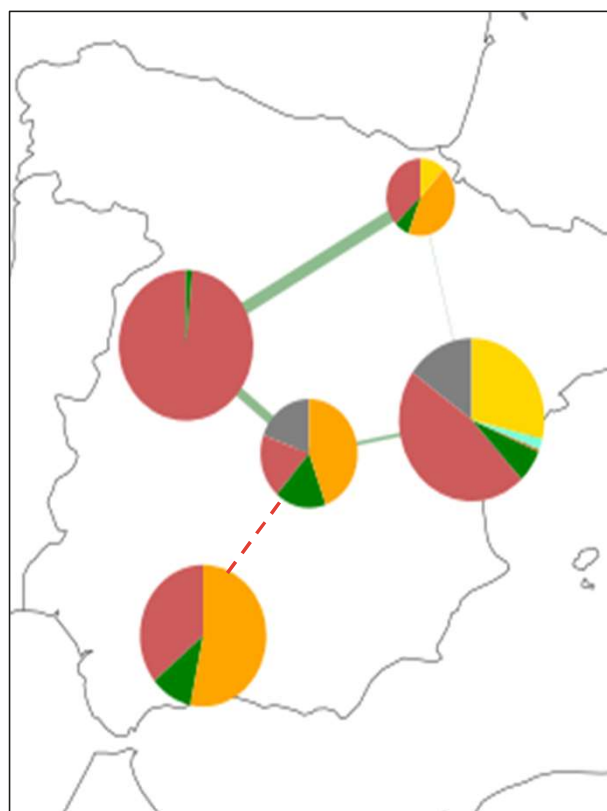
Dispatch Results

No CO₂ emissions and Purely RES

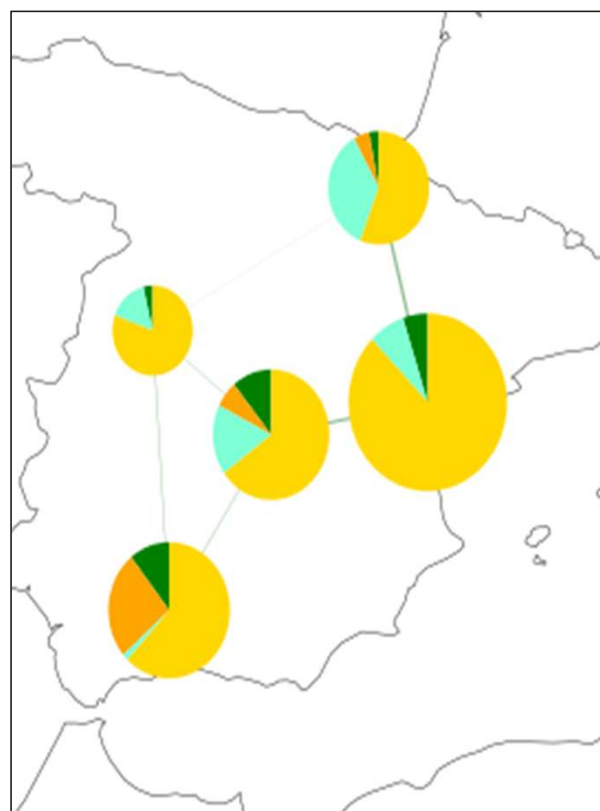


Transmission and Bus Capacities

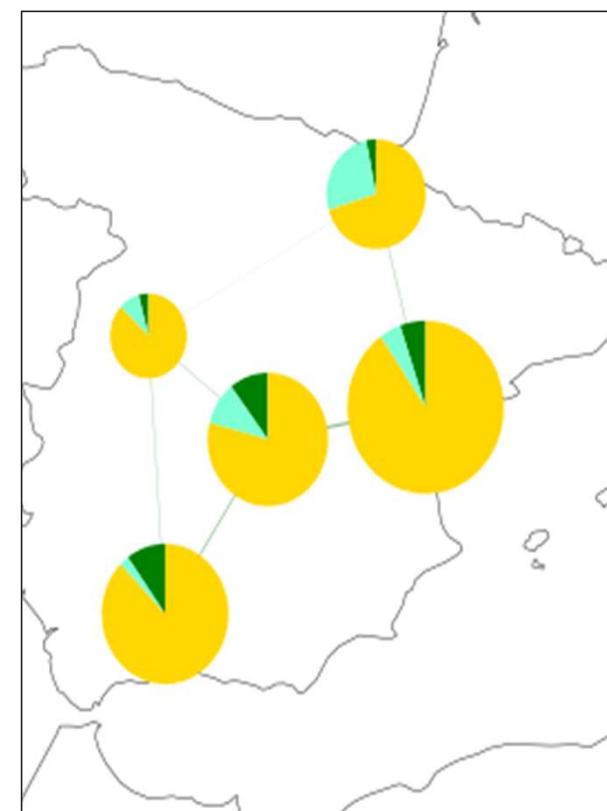
Base Scenario vs No CO₂ emissions and Purely RES



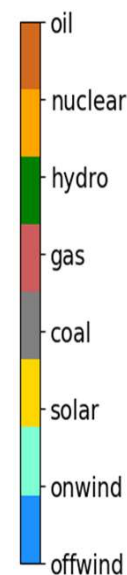
Base Scenario



No CO₂ emissions

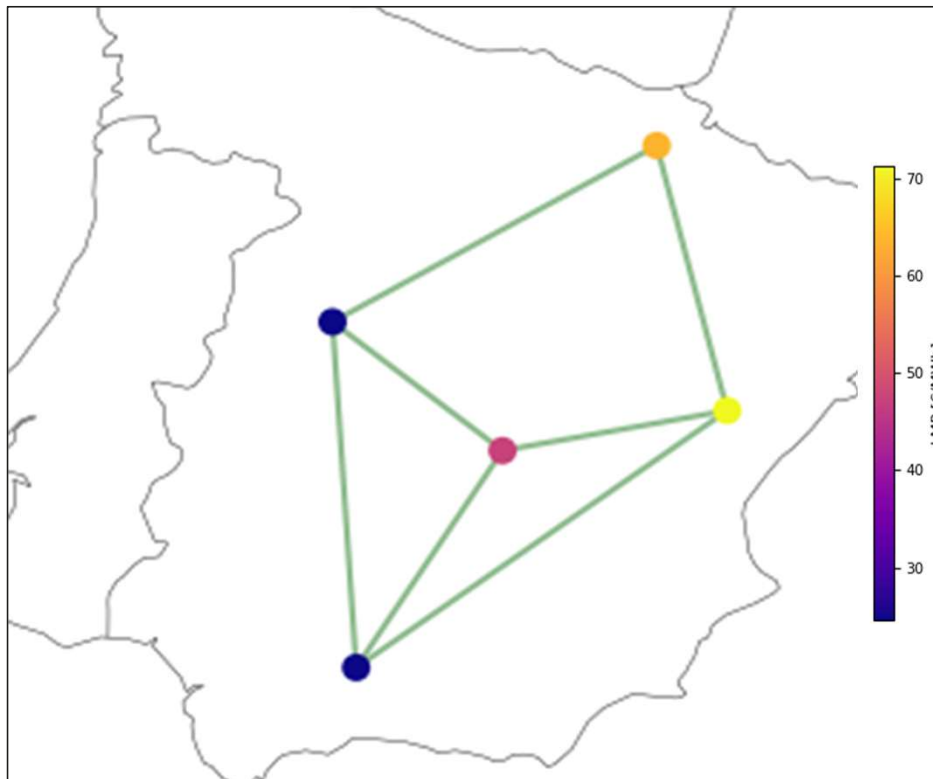


Purely RES

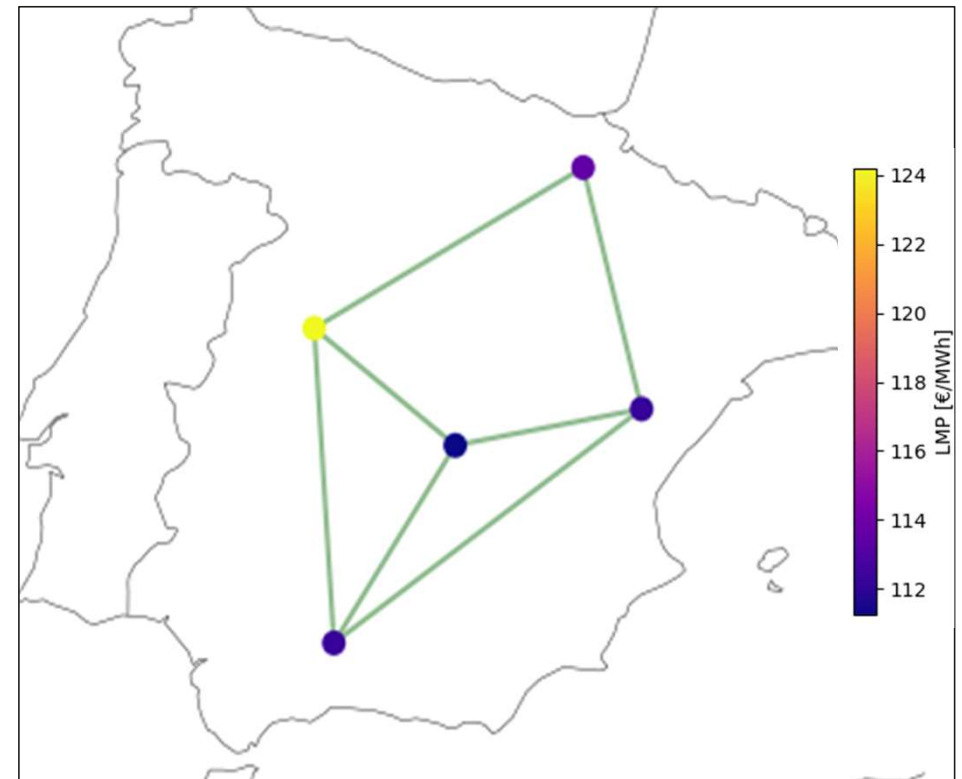


Marginal Price at Bus / Electricity Price

Base Scenario vs No CO₂ Emissions



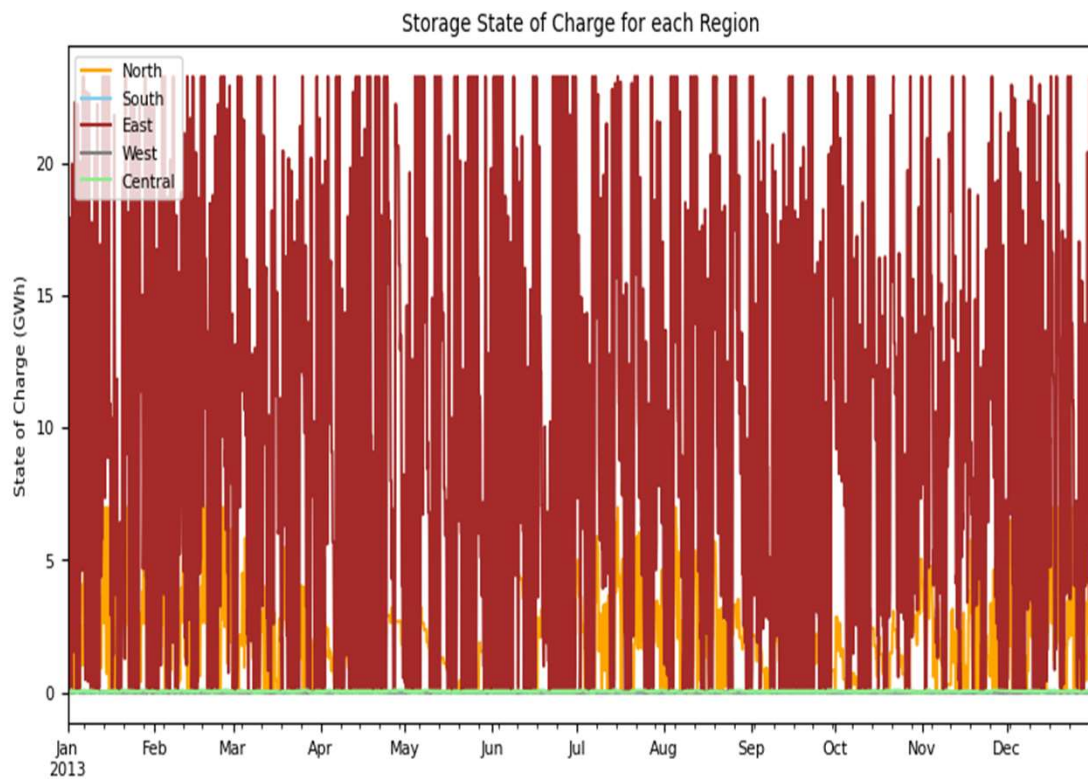
Base Scenario



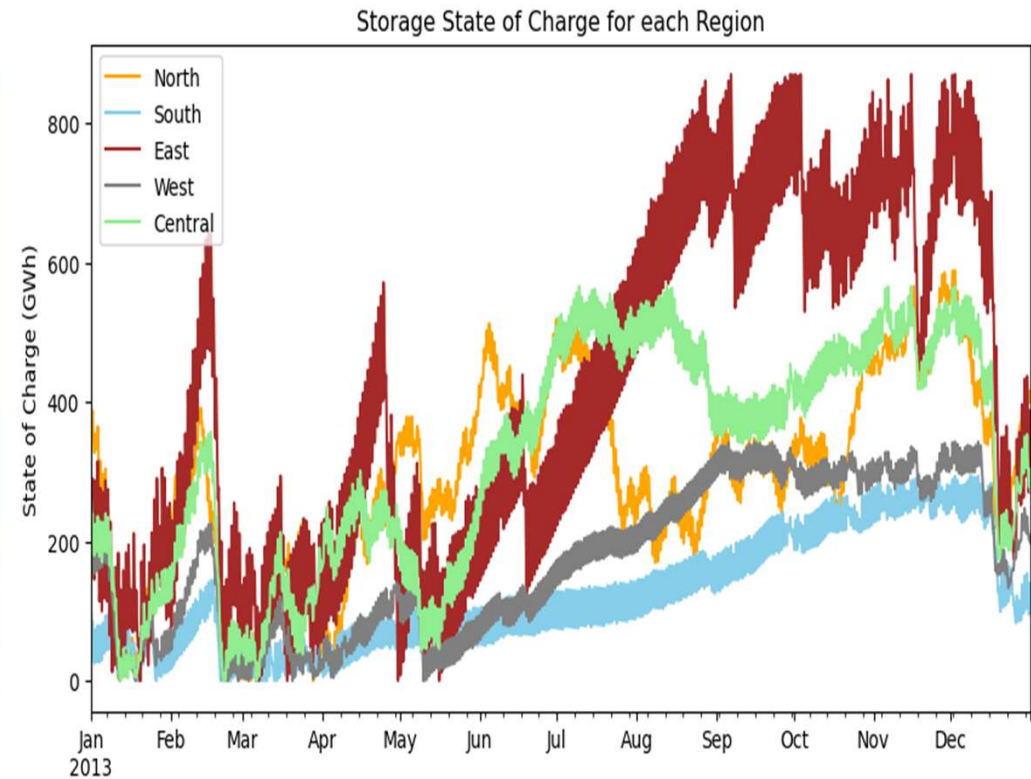
No CO₂ emissions

State of Charge

Base Scenario vs No CO₂ Emissions



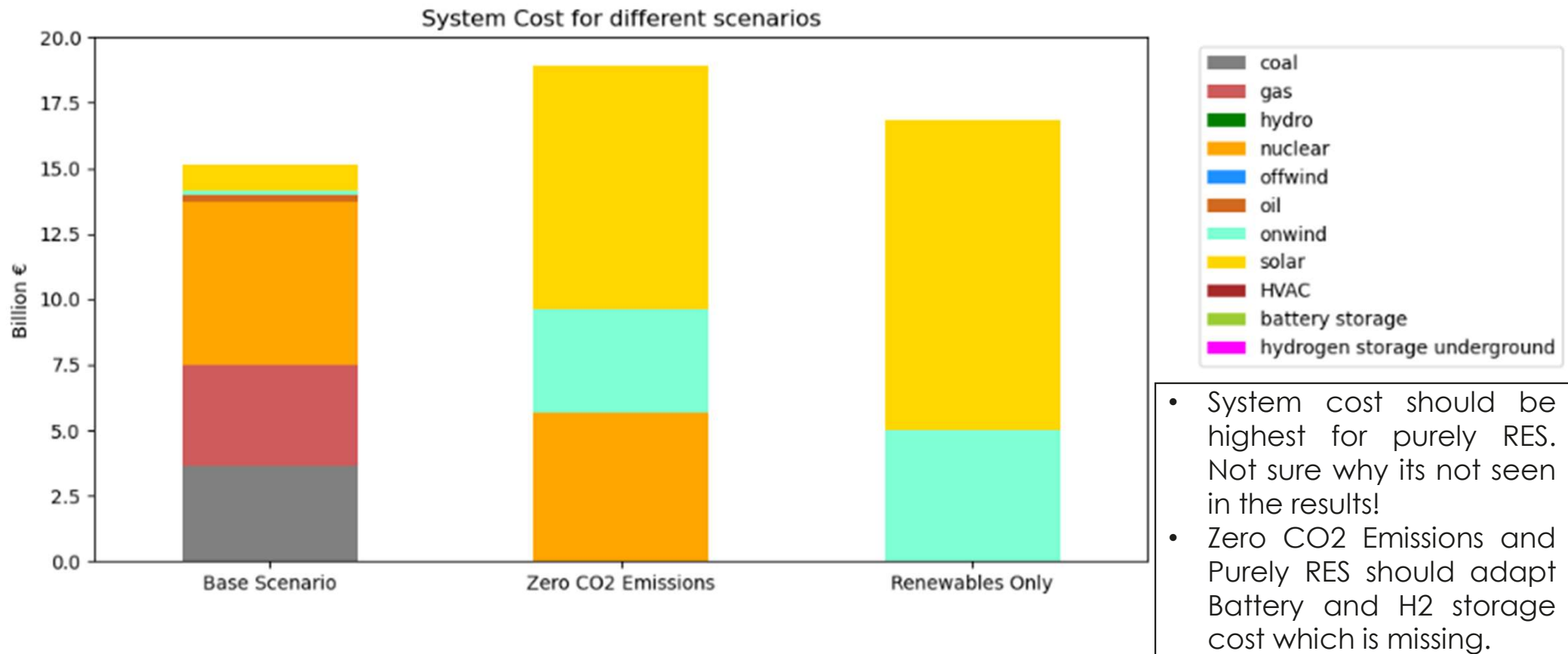
Base Scenario



No CO₂ emissions

System Cost

Base Scenario vs No CO₂ emissions and Purely RES



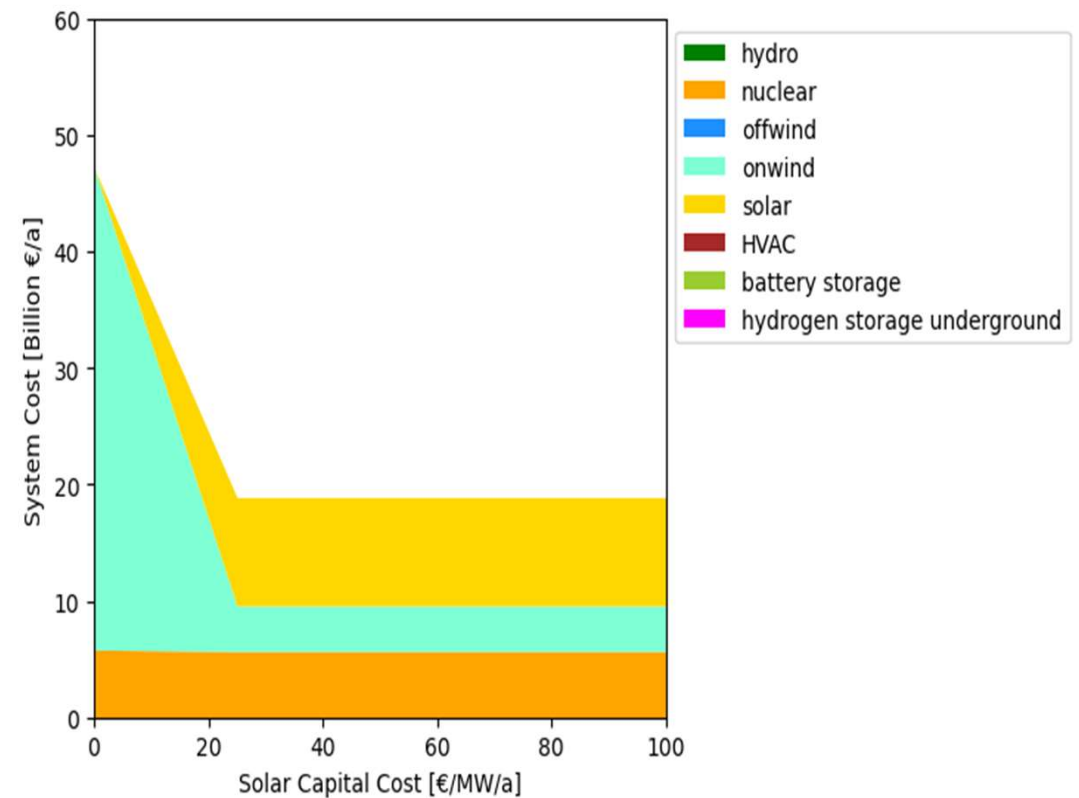
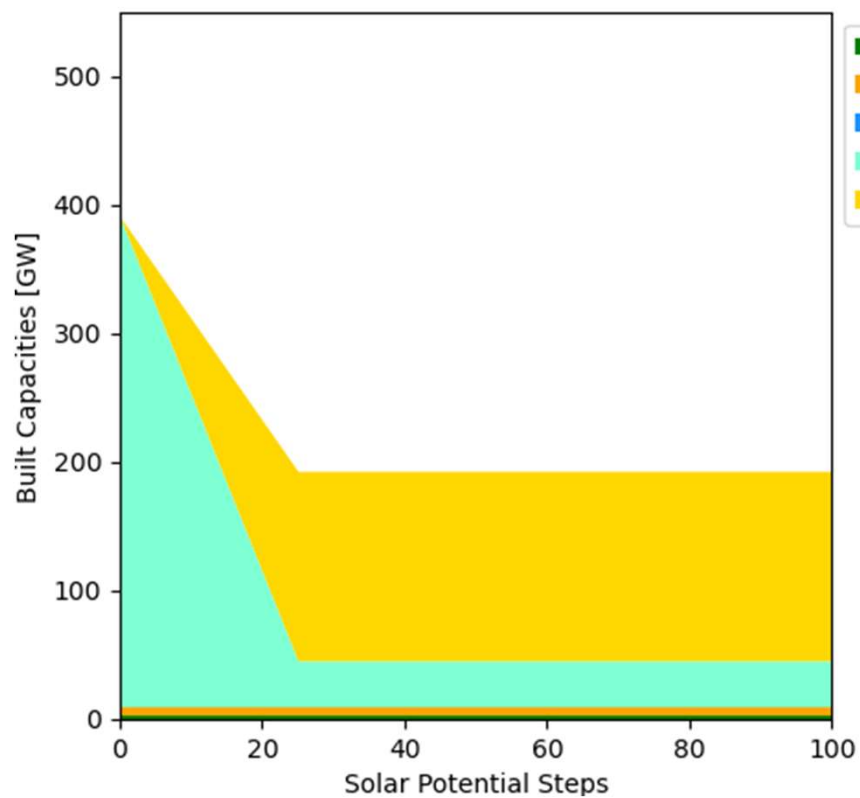
** No errors/bugs found within the code, still the output has is not accurate.

SENSITIVITY ANALYSIS

- Solar Dispatch and Cost Sensitivity

Solar Capacities and System Cost

Sensitivity Analysis



Conclusions

- The Spanish energy mix has high traces of '**Gas**' nearly 26GW.
- The country has **good potential** to install more **Onshore wind and Solar PV**, but on the other hand has **less scope with Offshore** wind (Referring to -50m depth).
- The **Northern region** is highly suitable for Onshore wind due to high wind speed and capacity factor, while the **Southern region** is highly suitable for Solar PV due to high horizontal irradiation.
- **High battery and hydrogen storage** will be needed to avoid any curtailment or less demands if entire system is running of purely renewables.
- Focusing entirely on renewables could allow for large-scale deployment and **the resulting economies of scale, driving down costs**.

Future Outlook

National Hydrogen Roadmap^[1]

- Approved by the Ministry for Ecological Transition and the Demographic Challenge (MITECO) in October 2020.
- Spain aims to replace 25% of the current 500,000 tonnes of fossil-based hydrogen consumed by the Spanish industry annually, with green hydrogen.
- The total electrolyser capacity target of Spain is 600 MW by end 2024, increasing to 4 GW by 2030.

Spanish NECP Targets (2021-2030)^[2]

- 23% reduction in GHG gas emissions.
- Nearly 42% renewable energy for end use.
- Nearly 42% improvement in the energy efficiency.
- 74% share of renewable electricity in generation

1. <https://www.trade.gov/market-intelligence/spain-renewable-hydrogen-roadmap>

2. <https://www.iea.org/reports/spain-2021>

‘Thank you’ – ‘Danke’ – ‘Gracias’

धन्यवाद!